IoT 메카트로닉스 (MES311)
2020-2

- Lecture 2 : System Modeling

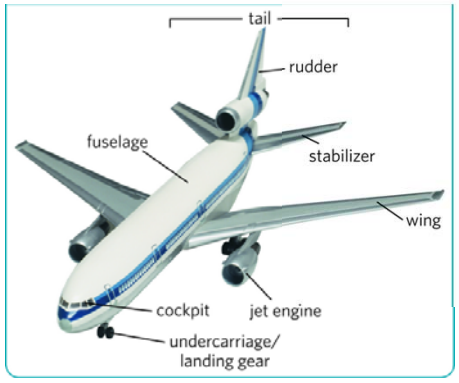
System

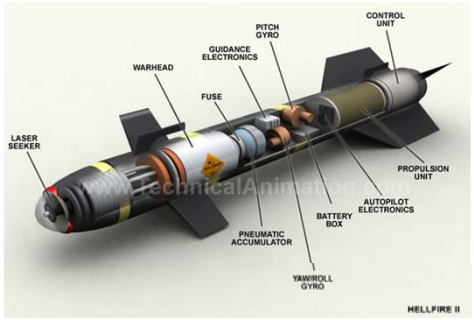
- System?
  - 각 구성요소들이 상호작용하거나 상호 의존하여 복잡하게 얽힌 통일된 하나의 집합체
  - 공학적의 의미의 시스템 예
    - 가전제품: 냉장고, 세탁기, 청소기,에어컨
    - 자동차, 로봇, 비행기, 미사일, 카메라









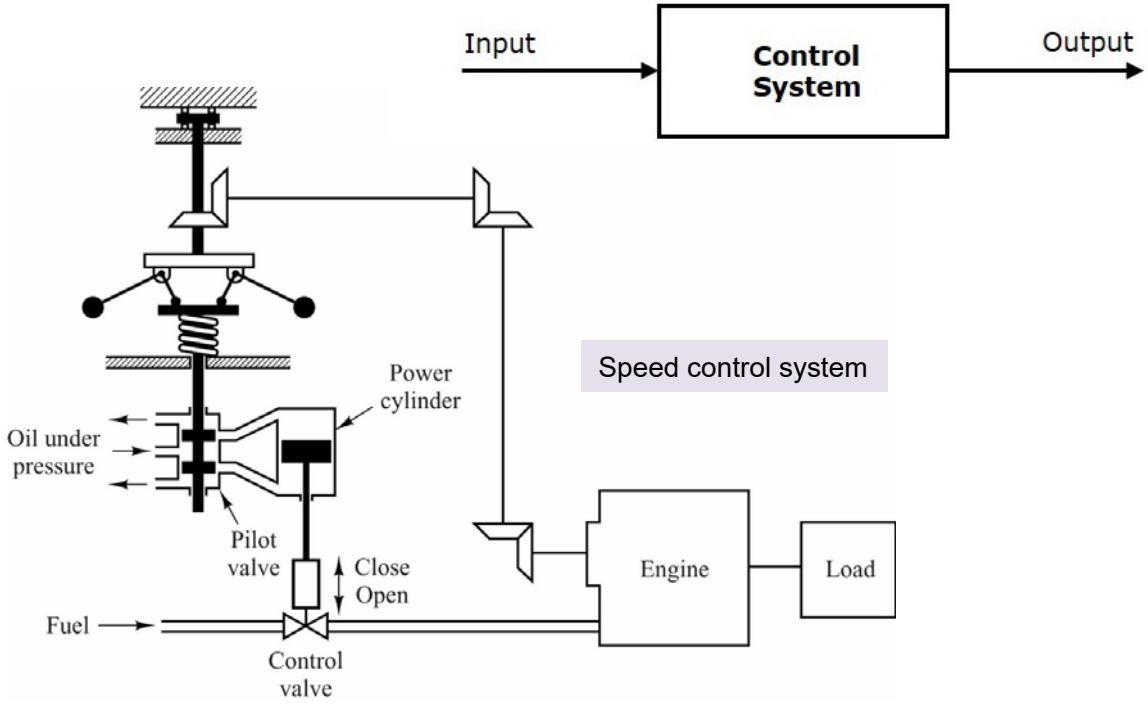




Control

- Control
  - 어떤 장치나 공정의 출력신호가 원하는 상태를 따라가도록 입력신호를 적절히 조절하는 것

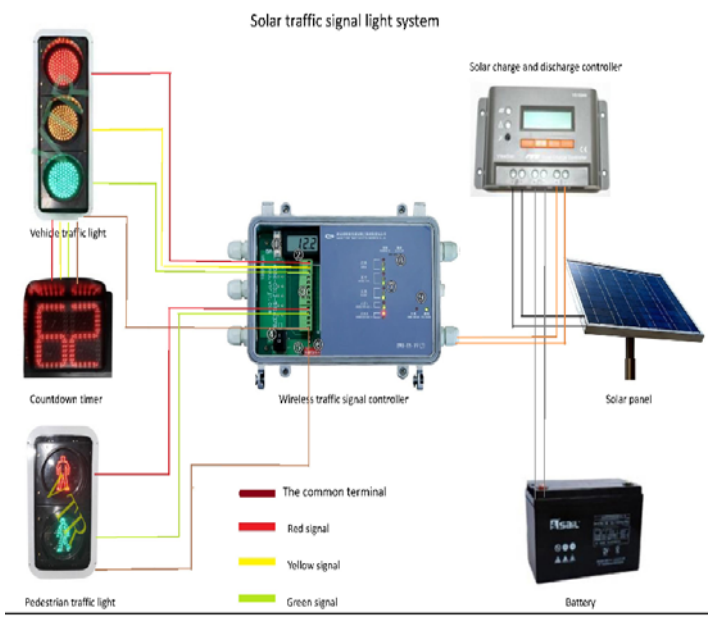
- Control System
  - 여러 부속품들이 상호 연결되어 출력이 입력을 추종하는 하는 시스템

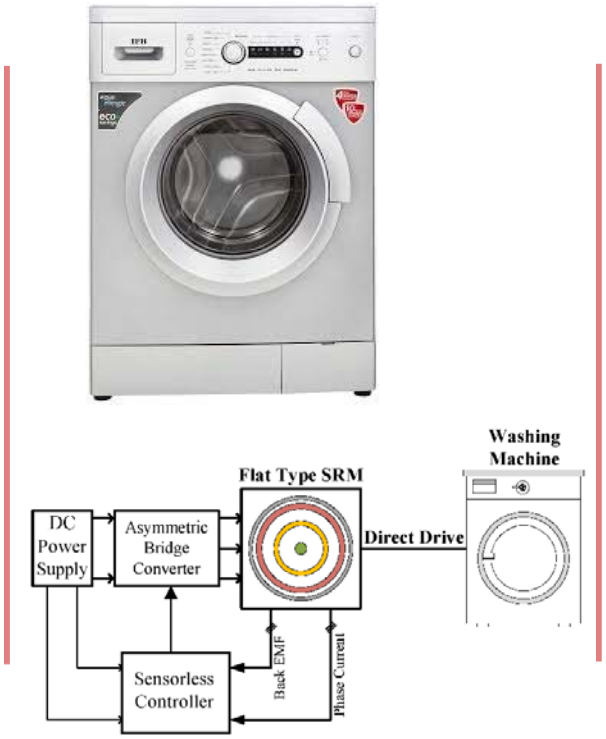


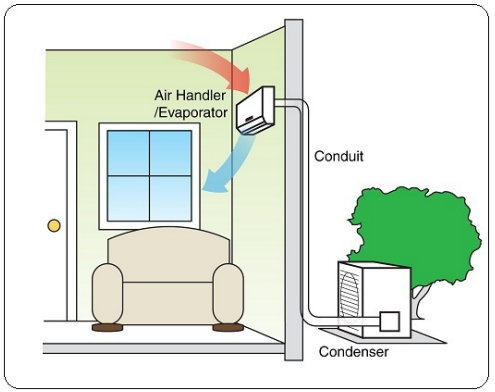
Speed control system

Control System

- Control system Examples
  - Traffic lights control system
  - Washing machine
  - Air conditioning system

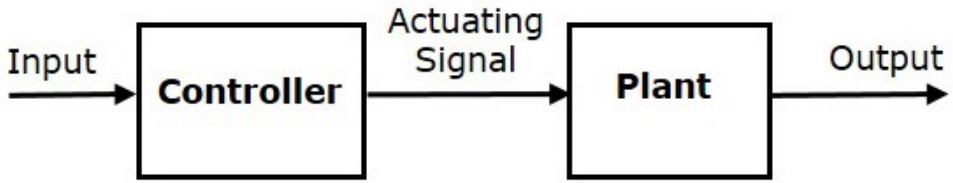




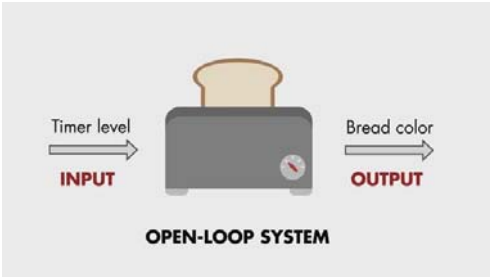


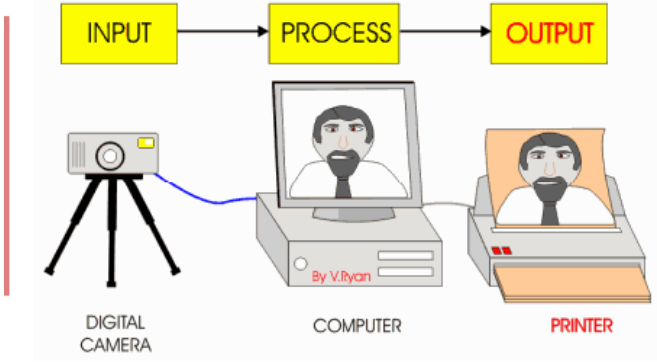
Control System

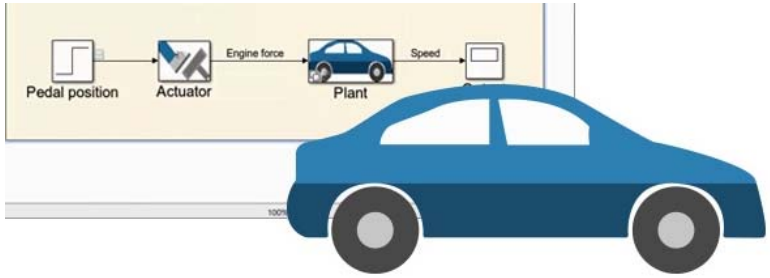
- Open-Loop Control System (Non-feedback Control system)
  - System 의 출력은 입력에 따라 변동하나 제어신호는 시스템 출력과는 무관



  - Examples

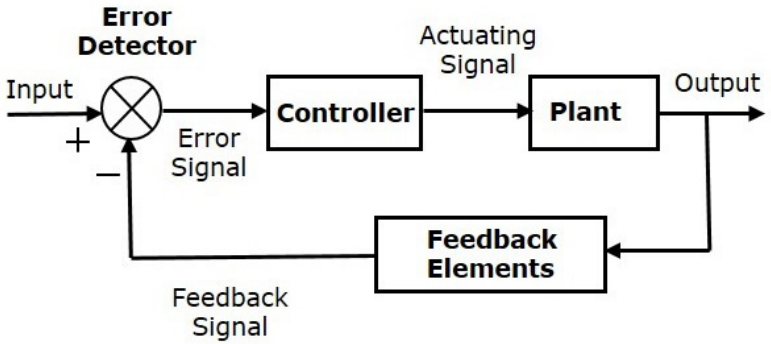




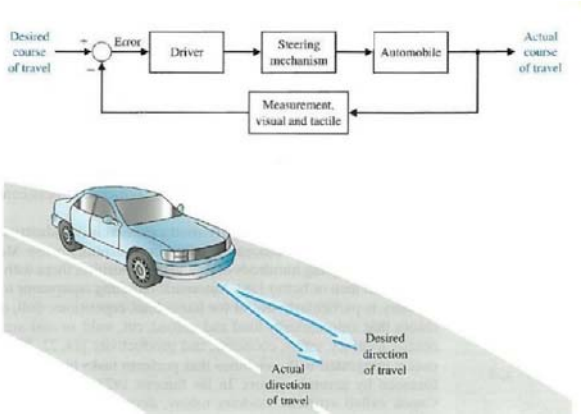


Control System

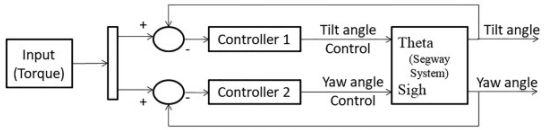
- Closed-Loop Control System (feedback Control system)
  - System 의 출력은 입력에 따라 변동하나 제어신호는 시스템 출력에 따라 변동

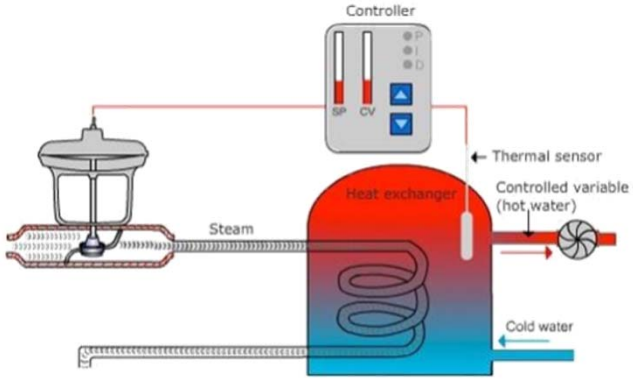


  - Examples

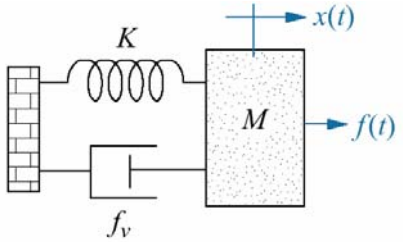


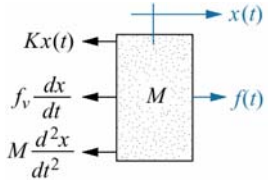




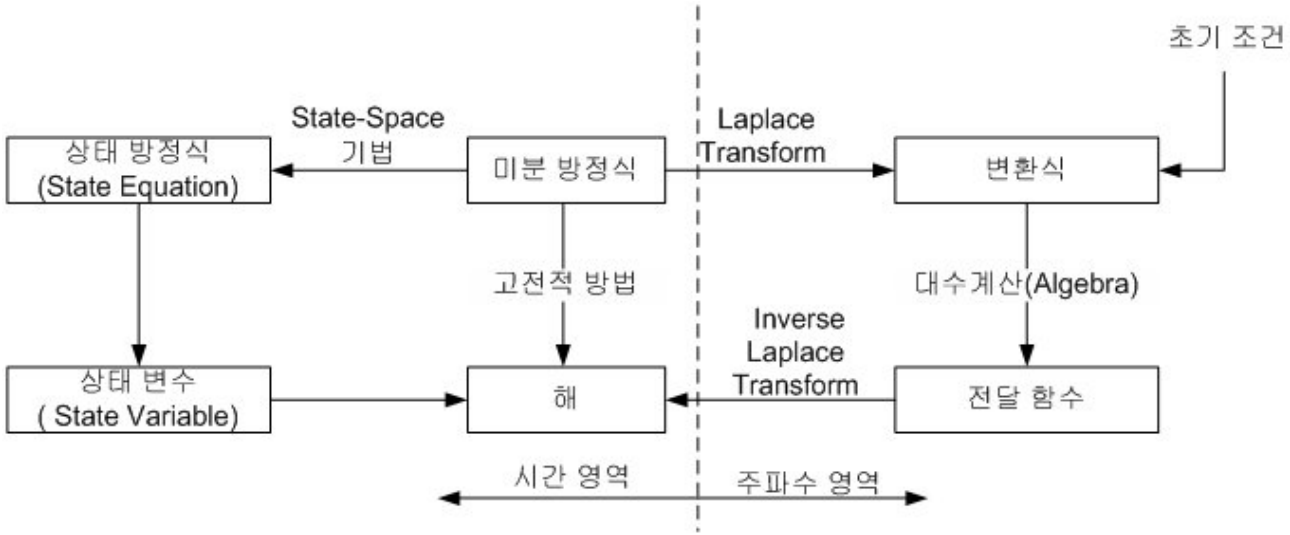


System Analysis



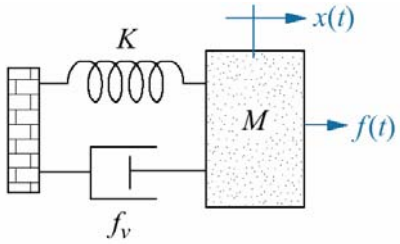


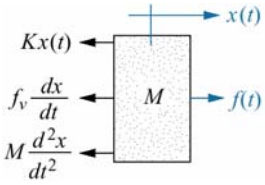
수학적 모델(운동방정식): $M\frac{d^2x(t)}{dt^2} + f_v\frac{dx(t)}{dt} + Kx(t) = f(t)$



Laplace Transform을 이용하면 시간영역에서 해를 구하기어려운 미분방정식을 보다 간단한 대수방정식으로 해를 구할 수 있다.

System Analysis example





수학적 모델(운동방정식): $M\frac{d^2x(t)}{dt^2} + f_v\frac{dx(t)}{dt} + Kx(t) = f(t)$

$State - Space - Equation$

$x_1 = y(t)$

$\dot{x}_1 = x_2$

$\dot{x}_2 = -\frac{f_v}{M}x_2 - \frac{K}{M}x_1 + \frac{f(t)}{M}$

$$\begin{bmatrix}\dot{x}_1 \\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1 \\ -\frac{K}{M} & -\frac{f_v}{M}\end{bmatrix}\begin{bmatrix}x_1 \\ x_2\end{bmatrix} + \begin{bmatrix}0 \\ \frac{1}{M}\end{bmatrix}f(t)$$

$$y = [1 \quad 0]\begin{bmatrix}x_1 \\ x_2\end{bmatrix}$$

적용대상: Linear, Non-linear,

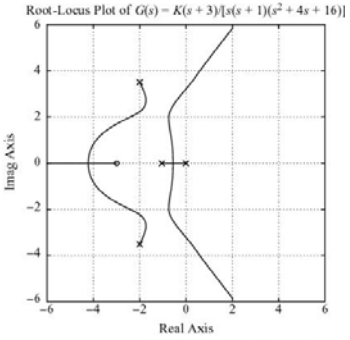
time-variant, time-invariant system

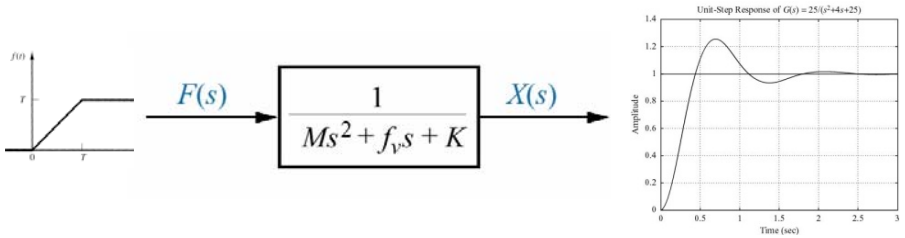
장점: 복잡한 시스템에 적합

Laplace변환:

$Ms^2X(s) + f_vsX(s) + KX(s) = F(s)$

$\frac{X(s)}{F(s)} = \frac{1}{Ms^2 + f_vs + K}$





적용대상: Linear, time-invariant system

장점: 안정도, 과도응답을 빨리 얻을 수 있다

직관적임

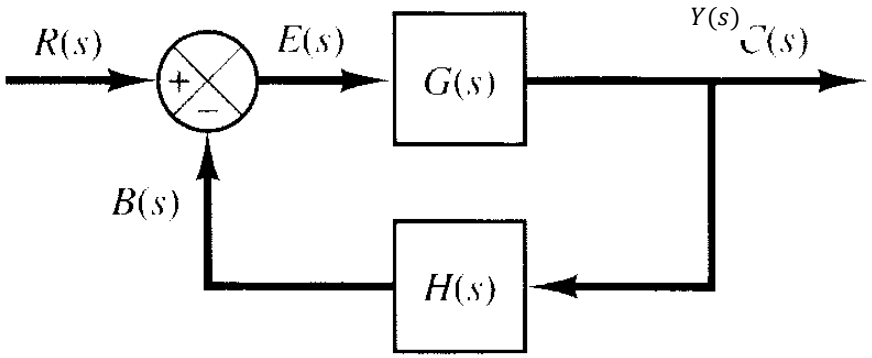
Laplace Transform theorem

- Laplace transform
  - 어떤 함수 f(t)에서 다른 함수로 변환하여 선형 동력계와 같은 미분방적식에 유용하게 사용
  - definition; $$F(s) = \mathcal{L}\{f\}(s) = \int_{0^-}^{\infty} e^{-st} f(t)\, dt.$$

- Few important theorem
  - 선형정리 (Linearity theorem): $$\mathcal{L}[kf(t)] = kF(s)$$ $$\mathcal{L}[f_1(t) + f_2(t)] = F_1(s) + F_2(s)$$
  - 미분정리(differentiation theorem): $$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0-)$$ $$\mathcal{L}\left[\frac{d^2f}{dt^2}\right] = s^2F(s) - sf(0-) - \dot{f}(0-)$$
  - 적분정리(Integration theorem): $$\mathcal{L}\left[\int_0^t f(\tau)\, d\tau\right] = \frac{F(s)}{s}$$

Transfer Function

- Transfer function:



$$Y(s) = E(s)G(s) \leftarrow E(s) = R(s) - B(s)$$

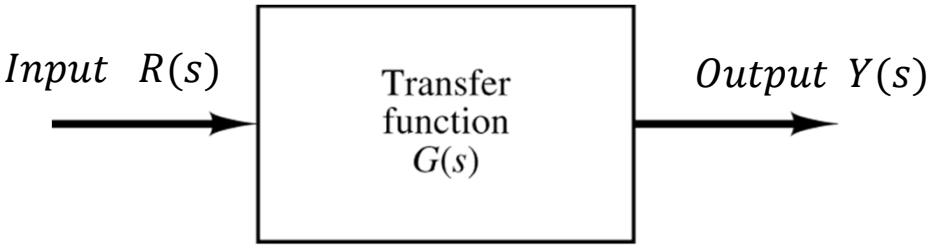
$$Y(s) = [R(s) - B(s)]G(s) \leftarrow B(s) = Y(s)H(s)$$

$$Y(s) = [R(s) - (Y(s)H(s))]G(s)$$

$$Y(s) = R(s)G(s) - Y(s)H(s)G(s)$$

$$Y(s) + G(s)H(s)Y(s) = R(s)G(s)$$

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

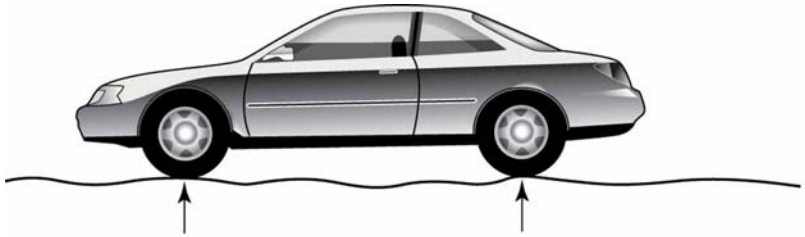


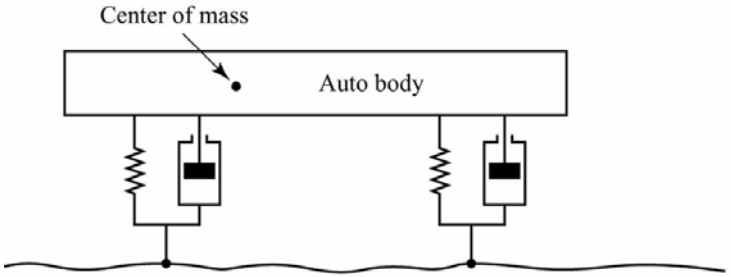
$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

Break time

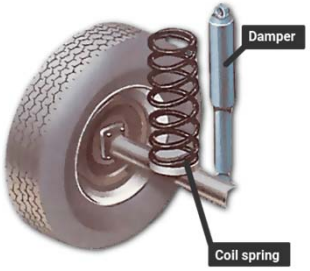
Modelling of Mechanical System

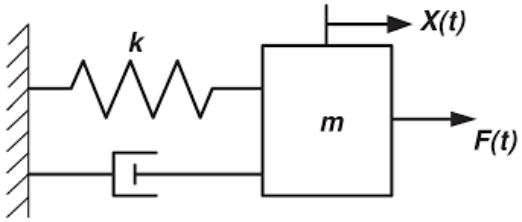
- Automobile suspension system





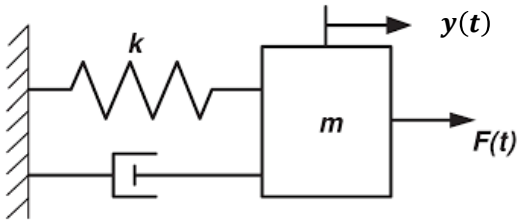
- Spring-Damper

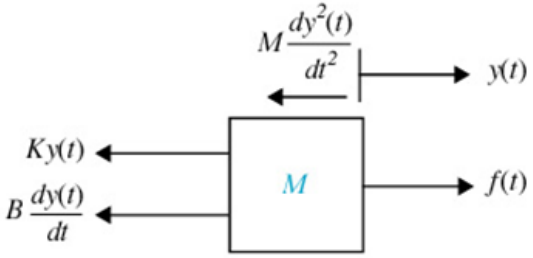




Modelling of Mechanical System

- Modeling of Mechanical System





Newton's 2nd Law

$$\sum F = Ma = M\ddot{y}(t)$$

$$\sum F = -Ky(t) - B\frac{dy}{dt} + f(t)$$

$$-Ky(t) - B\frac{dy}{dt} + f(t) = M\ddot{y}(t)$$

Equation of Motion

$$M\ddot{y}(t) + B\frac{dy}{dt} + Ky(t) = f(t)$$

- Laplace Transform

$$Ms^2Y(s) + BsY(s) + KY(s) = F(s)$$
$$\Rightarrow \frac{Y(s)}{F(s)} = \frac{1}{Ms^2 + Bs + K}$$

Modelling of Electrical Componets

- Capacitor



$$\frac{1}{C}\int idt = v(t)$$ C: unit: F(Farad))

  - Impedance: $Z(s) = \frac{V(s)}{I(s)} = \frac{1}{Cs}$

- Resistor



$$Ri = v(t)$$ R: unit: Ω (Ohm)

  - Impedance: $Z(s) = \frac{V(s)}{I(s)} = R$

- Inductor

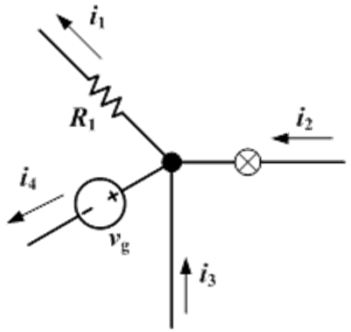


$$L\frac{di}{dt} = v(t)$$ L: unit: H (Henry)

  - Impedance: $Z(s) = \frac{V(s)}{I(s)} = Ls$

Kirchhoff's law

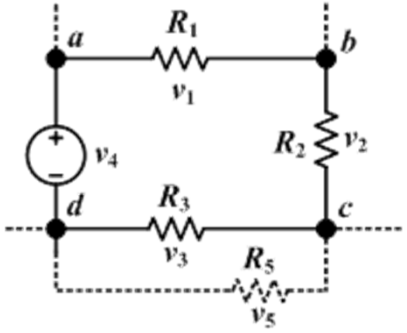
- **Kirchhoff's current law (KCL)**
  - 마디에 흐르는 전류의 합은 0이다



$$\sum_{k=1}^{n} I_k = 0$$

$$-i_1 + i_2 - i_4 + i_3 = 0$$

- **Kirchhoff's voltage law (KVL)**
  - 회로망속의 임의의 폐회로(閉回路, Closed Circuit)안에서 폐루프를 일주하면서 취한 전압의 합은 0이다

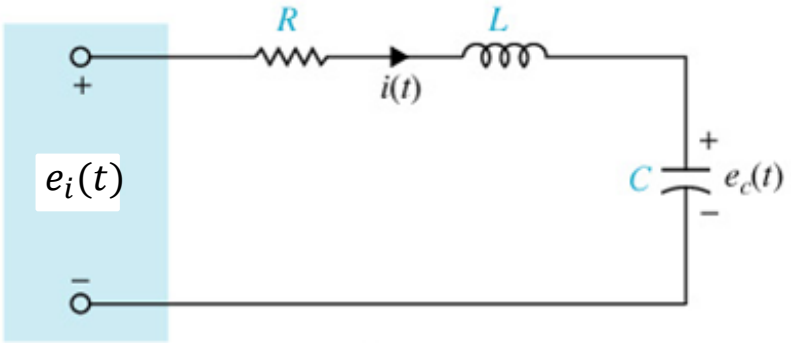


$$\sum_{k=1}^{n} V_k = 0$$

$$V_4 - V_1 - V_2 - V_3 = 0 \Rightarrow V_4 = V_1 + V_2 + V_3$$

Modelling of Electrical System

- LRC Circuit



$$Ri + L\frac{di}{dt} + \frac{1}{C}\int idt = e_i(t)$$

$$\frac{1}{C}\int idt = e_c(t)$$

– Laplace Transform

$$LsI(s) + RI(s) + \frac{1}{C}\frac{1}{s}I(s) = E_i(s)$$

$$\frac{1}{C}\frac{1}{s}I(s) = E_o(s)$$

$$\Rightarrow \quad \frac{E_o(s)}{E_i(s)} = \frac{1}{LCs^2 + RCs + 1}$$

Impedance

- Impedance?
  - 부가된 전류에 대한 전압의 비 $Z = \frac{V(t)}{I(t)}, \frac{V(s)}{I(s)}$

- 직렬연결

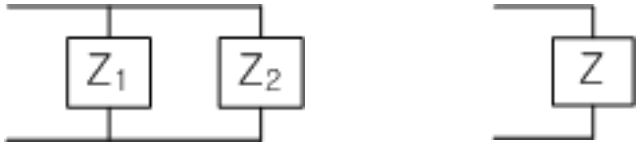
$$Z = Z_1 + Z_2$$



- 병열연결

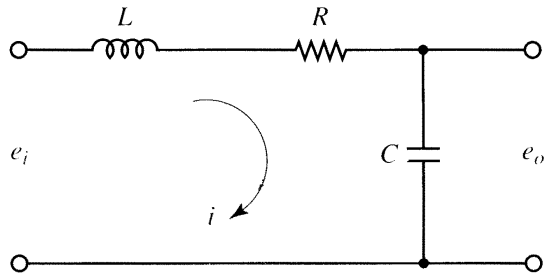
$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$Z = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

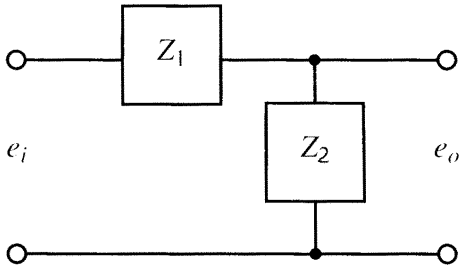


Impedance을 이용한 전기적 모델

- LRC Circuit



simplify



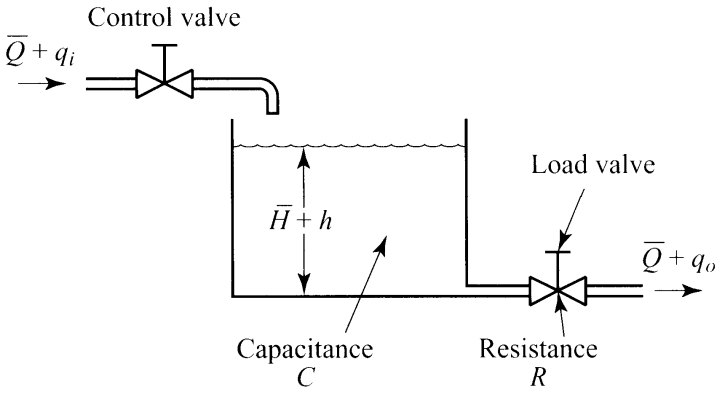
$$Z_1 = Ls + R$$

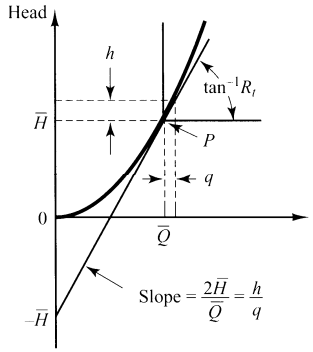
$$Z_2 = \frac{1}{Cs}$$

- Transfer function

$$\frac{E_o(s)}{E_i(s)} = \frac{Z_2}{Z_1 + Z_2} = \frac{\frac{1}{Cs}}{Ls + R + \frac{1}{Cs}} = \frac{1}{LCs^2 + RCs + 1}$$

Liquid Level System





$\bar{Q}$ = 정상상태의 유량, m^3/sec

q_i = 정상상태에서의 유입량의 편차, m^3/sec

q_o = 정상상태에서의 유출량의 편차, m^3/sec

$\bar{H}$ = 정상상태의 수두, m

h_i = 정상상태에서의 수두의 편차, m

유량과 수두의 관계식

– Laminar flow: $Q = KH$,

– Turbulence flow: $Q = K\sqrt{H}$

유체흐름에 대한 저항: $R = \frac{\text{액위차 변화량}(m)}{\text{유량변화량}(flowrate)(m^3/\text{sec})}$

– Laminar flow: $R_l = \frac{dH}{dQ} = \frac{H}{Q}$, – Turbulence flow: $R_t = \frac{dH}{dQ} = \frac{2\sqrt{H}}{K} = \frac{2H}{Q}$

탱크의 커패시턴스: $C = \frac{\text{저장액체 변화량}(m^3)}{\text{수두변화량}(m)}$

- 탱크에 축적된 량 = dt동안 유입량과 유출량의 차

$$Cdh = (q_i - q_o)dt \Rightarrow RC\frac{dh}{dt} + h = Rq_i$$

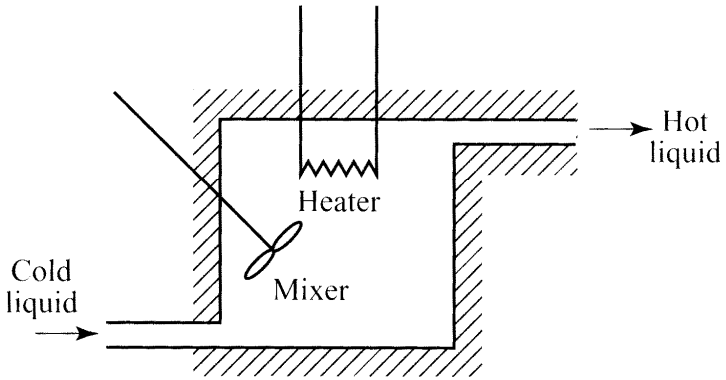
- Laplace Transform:

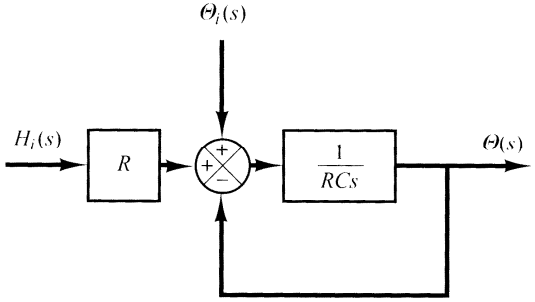
$$(RCs + 1)H(s) = RQ_i(s)$$

- Transfer Function:

$$\frac{H(s)}{Q_i(s)} = \frac{R}{RCs + 1} \quad \frac{Q_o(s)}{Q_i(s)} = \frac{1}{RCs + 1} = \frac{1}{Ts + 1}, T = RC$$

Thermal System





$\bar{\Theta}_i$ = 유입액체의 정상상태의 온도 $°C$

$\bar{\Theta}_o$ = 유출액체의 정상상태의 온도 $°C$

G = 정상상태의 유량, $kg/\sec$

M = 탱크내의 액체 질량, kg

c = 액체의 비열, $kcal/kg°C$

R = 열저항, $°C \sec/ k\ cal$

C = 열커패시턴스, $kcal/°C$

$\bar{H}$ = 정상상태 열유입률, $kcal/sec$

h_i = 정상상태에서의 열유입률 변화량

h_o = 정상상태에서의 열유출률 변화량

• 정상상태에서의 열 시스템의 변화

$h_o = G \times c \times \theta$

$C = M \times c$

$R = \frac{\theta}{h_o} = \frac{1}{G \times c}$

• 열 평형식: $Cd\theta = (h_i - h_o)dt$

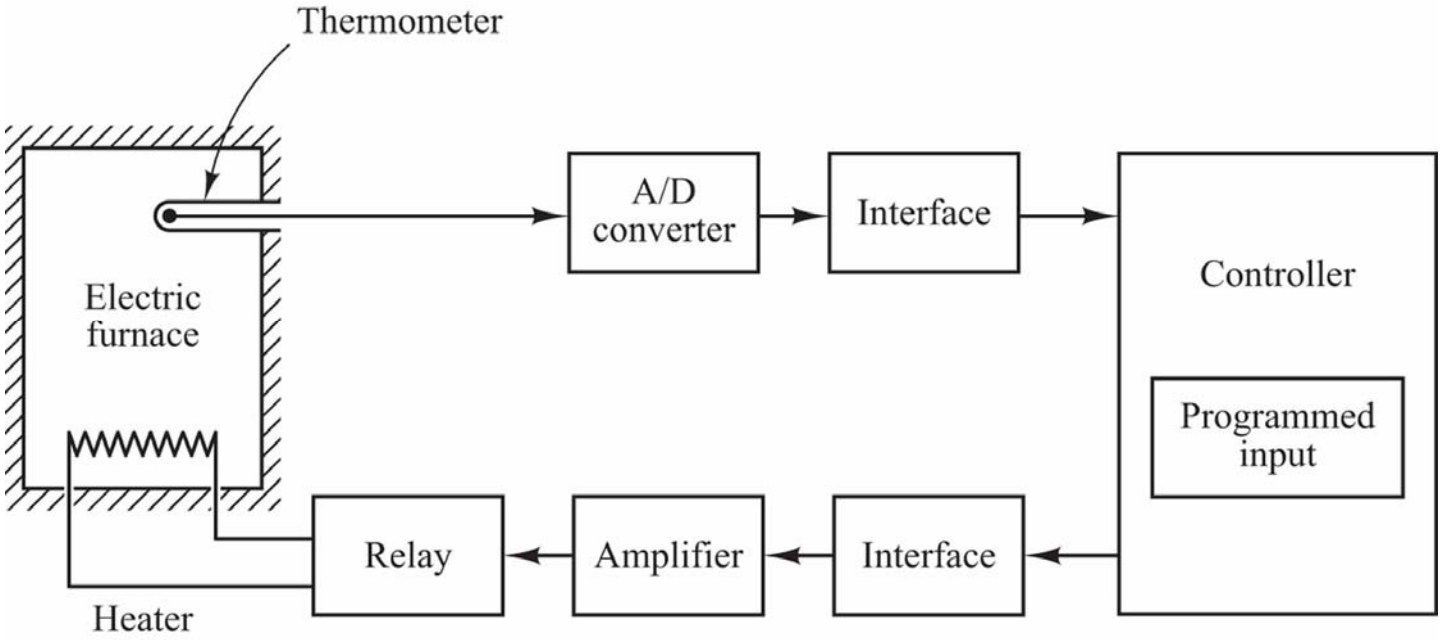
$$C\frac{d\theta}{dt} = h_i - h_o, h_o = \frac{1}{R}\theta$$

$$\Rightarrow RC\frac{d\theta}{dt} + \theta = Rh_i$$

• 전달함수

$$\frac{\Theta(s)}{H_i(s)} = \frac{R}{RCs+1}$$

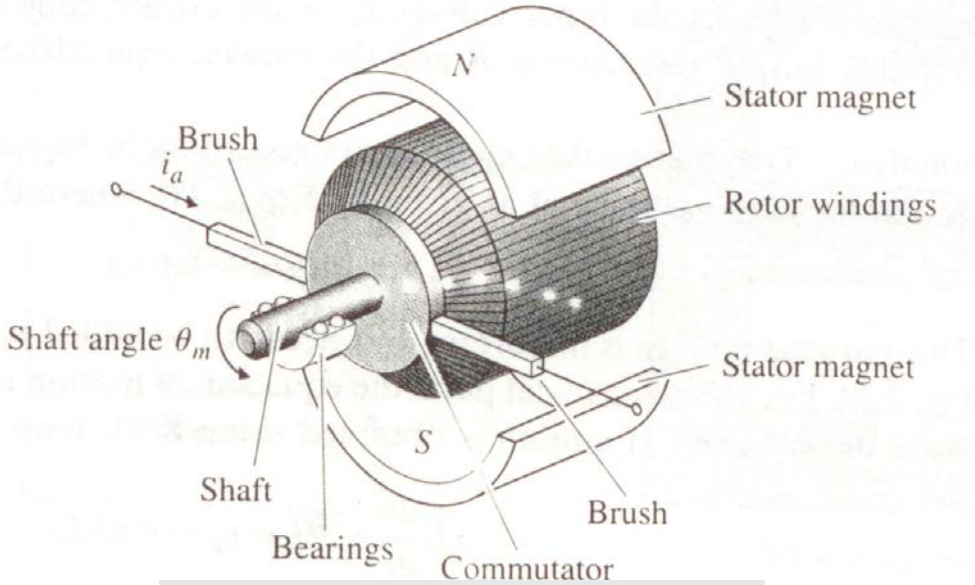
Thermal Control System



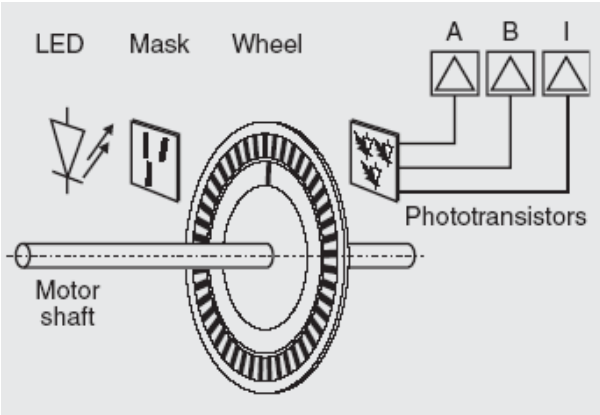
DC Motor

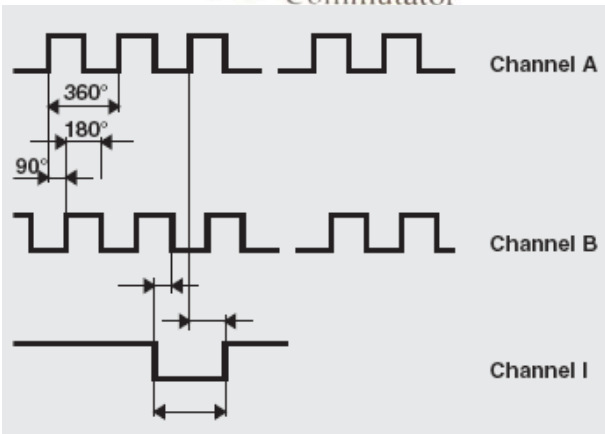
- DC Motor 구조



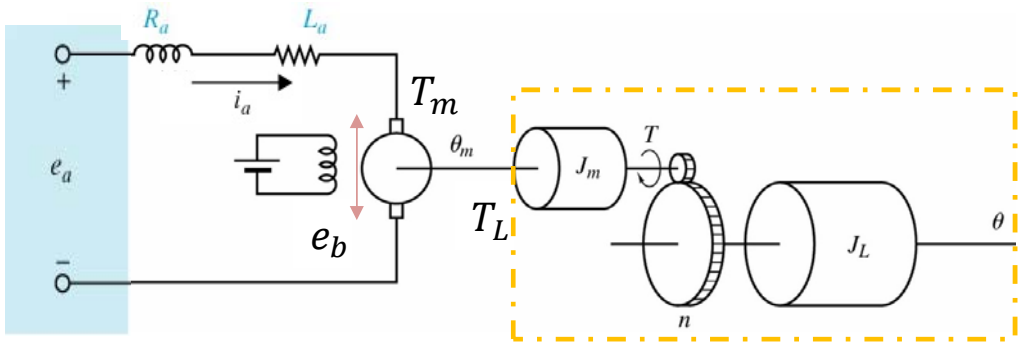


- Opto-electric Encoder





Modelling of DC Motor



$$\sum V = 0 \Rightarrow e_a = R_a i + L_a \frac{di}{dt} + e_b$$

$$\sum M = J\alpha \Rightarrow -B_m\dot{\theta}_m + T_m - T_L = J\ddot{\theta}_m$$

$$\Rightarrow \frac{\theta_m(s)}{E_a(s)} = \frac{K_i}{L_a J s^3 + (R_a J + B_m L_a)s^2 + K_i K_b + R_a B_m s}$$

$T_m = K_i i_a(t), e_b = K_b\dot{\theta}_m$

K_i : $Torque$ Constant (Nm/A)

K_b : $BackElectricmotiveForce(BackEMF)$ 상수

L_a : 전 기자 인덕턴스

R_a : 전 기자 저항

e_a : 인 가전압

e_b : 역 기전력

Block Diagram of DC Motor System

